Proof of Theorem 2.1

Continuous compounding as a limit, step by step

> **Theorem 2.1.** For every $P > 0, r > 0, t > 0$:
>
> $$\lim_{n\to\infty} P\left(1+\frac{r}{n}\right)^{nt} = Pe^{rt}.$$

Plan: (1) take a logarithm, (2) bound the logarithm, (3) squeeze, (4) use that **exp** is continuous.

Step 0. Why we cannot just plug in $n = \infty$

As $n \to \infty$ the base $1 + \frac{r}{n} \to 1$, but the exponent $nt \to \infty$. This is the form 1^∞, which is **indeterminate**. A base close to 1 pulls the value down toward 1, and a huge exponent pushes it up, so the answer depends on which effect wins. So the answer is **not** simply 1. We need a method.

Step 1. Turn the power into a product

For any $a > 0$ we have $a^b = e^{b \ln a}$. Here $a = 1 + \frac{r}{n} > 0$, so

$$A_n = P\left(1+\frac{r}{n}\right)^{nt} = Pe^{x_n}, \qquad \text{where} \qquad x_n = nt\,\ln\left(1+\frac{r}{n}\right).$$

Now the hard part is only the exponent x_n, which has the form $\infty \cdot 0$. **New question: what does x_n converge to?**

Step 2. The key inequality (Lemma 1.5)

> **Lemma.** For every $u > 0$:
>
> $$\frac{u}{1+u} \;<\; \ln(1+u) \;<\; u.$$

Proof. Write the logarithm as an integral:

$$\ln(1+u) = \int_0^u \frac{1}{1+s}\,ds.$$

For $0 < s < u$ the integrand is between two constants:

$$\frac{1}{1+u} \;<\; \frac{1}{1+s} \;<\; 1.$$

Integrate all three over $[0, u]$, an interval of length u:

$$\underbrace{\int_0^u \frac{1}{1+u}\,ds}_{=\frac{u}{1+u}} \;<\; \underbrace{\int_0^u \frac{1}{1+s}\,ds}_{=\ln(1+u)} \;<\; \underbrace{\int_0^u 1\,ds}_{=u}. \qquad \blacksquare$$

Picture: the area under the curve $1/(1+s)$ lies between a short rectangle (height $\frac{1}{1+u}$) and a tall one (height 1). The main guide proves the same lemma with $\psi(u) = u - \ln(1+u)$; that version also covers $-1 < u < 0$, which we don't need here.

Step 3. Apply the lemma and squeeze

Put $u = \frac{r}{n} > 0$ into the lemma and multiply everything by $nt > 0$ (a positive number does not change the direction of the inequalities):

$$nt \cdot \frac{r/n}{1 + r/n} \;<\; nt \ln\!\left(1 + \frac{r}{n}\right) \;<\; nt \cdot \frac{r}{n}.$$

Simplify both sides (the n cancels):

$$\frac{rt}{1 + \frac{r}{n}} \;<\; x_n \;<\; rt.$$

- **Right side:** the constant rt.
- **Left side:** $1 + \frac{r}{n} \to 1$, so $\dfrac{rt}{1 + r/n} \to \dfrac{rt}{1} = rt$ (quotient rule for limits; allowed because the denominator tends to $1 \neq 0$).

Both bounds tend to rt, so by the **squeeze theorem**: $x_n \to rt$.

The same step with ε–N. From the green box,

$$0 \;<\; rt - x_n \;<\; rt - \frac{rt}{1 + r/n} \;=\; rt \cdot \frac{r/n}{1 + r/n} \;<\; \frac{r^2 t}{n}.$$

Given $\varepsilon > 0$, choose $N > \dfrac{r^2 t}{\varepsilon}$. Then for all $n > N$: $|x_n - rt| < \dfrac{r^2 t}{n} < \varepsilon$. That is exactly the definition of $x_n \to rt$.

Step 4. Move the limit inside exp

The function e^x is continuous. By sequential continuity (Lemma 1.3), $x_n \to rt$ implies $e^{x_n} \to e^{rt}$. Therefore

$$\lim_{n\to\infty} A_n \;=\; P \lim_{n\to\infty} e^{x_n} \;=\; P\, e^{\lim x_n} \;=\; P\, e^{rt}. \qquad \blacksquare$$

Bonus. An explicit error bound from the same proof

Let $\varepsilon_n = rt - x_n$. By Step 3, $0 < \varepsilon_n < \frac{r^2 t}{n}$. Using $1 - e^{-y} \le y$:

$$0 \;<\; Pe^{rt} - A_n \;=\; Pe^{rt}\left(1 - e^{-\varepsilon_n}\right) \;\le\; Pe^{rt}\,\varepsilon_n \;<\; Pe^{rt}\,\frac{r^2 t}{n}.$$

So the gap shrinks like $\frac{1}{n}$. Check with the project numbers ($P = 10{,}000$, $r = 0.1$, $t = 10$, daily: $n = 365$):

$$\text{bound} = 27{,}182.82 \cdot \frac{0.01 \cdot 10}{365} \approx 7.45, \qquad \text{actual gap} = 3.72 \;\checkmark$$

The actual gap is half the bound. The Taylor expansion in §2.3 of the guide gives the sharper constant $\frac{r^2 t}{2n}$; that factor $\frac{1}{2}$ explains it.

Where each assumption was used

Assumption	Used in
$1 + \frac{r}{n} > 0$	Step 1: $a^b = e^{b \ln a}$ needs $a > 0$
$r > 0$	Step 2: lemma with $u > 0$
$nt > 0$	Step 3: multiplying keeps the inequality directions
$\exp$ is continuous	Step 4: moving the limit inside

Check yourself. Use the green box with $r = 2, t = 1$ to show that $\lim_{n \to \infty} \left(1 + \frac{2}{n}\right)^n = e^2 \approx 7.389$.